Shresth Singh

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Portfolio Website: shresthsinghh-dotcom.github.io/portfolio

Education

Duke University — Durham, NC | B.S. Mechanical Engineering | Expected May 2029 | GPA: 4.0 / 4.0 | ACT 36

Relevant Coursework: Linear Algebra | Academic Activities Duke Math Meet participant (2024, 2025)

Quantitative, Computational & Engineering Experience

Structural & Aerodynamic Engineer (Modeling, Validation, & Systems) | Duke AERO 2025 – Present

- Developed and executed computational models for structural loading; analyzed subsystem behavior under varying boundary conditions using FEA (ANSYS) and interpreted outputs for insight, identifying potential failure points.
- Improved subsystem modeling accuracy to within 2% tolerance by validating FEA simulations against physical constraints.
- Coordinated cross functional integration of simulation outputs across component interfaces and aligned modeling to reduce rework and improve subsystem compatibility.

Professional Mathematics and Analysis Experience

Lead Mathematics & Physics Tutor | Apex Academix 2021 - Present

- Improved academic performance for 20+ students by implementing, data-tracked intervention plans in calculus and physics.
- Diagnosed underlying misconceptions and adapted explanations using algebraic, graphical, and physical representations.
- Tracked performance data via Excel to evaluate progress and iteratively refine solution strategies for standardized testing.

Data Analyst Intern | Relanto Summer 2024

- Used SQL to perform structured data validation ensuring relational dataset integrity, supporting data driven decisions.
- Reduced model deployment risk by identifying systematic failure modes through Python-based parameter testing.

Taekwondo Instructor | Vision Martial Arts 2023 - 2025

- Mentored students toward measurable progress using performance metrics, ensuring rank and milestone achievements.
- Increased instructional consistency and efficiency by standardizing class coordination workflows and implementing iterative performance improvements.

Academic Projects

Design & Analysis Team Member | Engineering 101 | Fall 2025

- Achieved 90% empirical system accuracy by modeling control logic and iteratively refining the prototype of an electronically operated launcher; designed end-to-end through residual analysis.
- Evaluated energy transfer and efficiency in a spring-actuated mechanism to inform performance optimization decisions.
- Executed quantitative design decisions using Onshape, Arduino, tolerancing principles, and technical memos.

Skills

Project & Process Skills: Project lifecycle coordination, Risk assessment, Process optimization, Technical documentation

Quantitative Modeling & Analysis: FEA (ANSYS), Engineering design, Exploratory data analysis, Statistical interpretation

Programming & Data Tools: Python (NumPy, Matplotlib), SQL, C++ (fundamentals), Excel, Git/GitLab, PowerPoint, Jupyter

Languages: English (Fluent), German (Intermediate), Hindi (Conversational)

Selected Activities

Technical Analyst and Public Speaker | Duke Cyber 2025 – Present

- Identified system vulnerabilities and failure risks by applying structured analysis in cross-functional security simulations.
- Applied formal network security models to evaluate system threats and mitigation strategies in simulations.

International Competitor and Public Speaker | DECA 2023 – 2025

- Secured State-Level award and was responsible for the sale of a house by a real estate agency by developing a quantitative market positioning strategy informed by competitive and demand analysis.

International Competitor and Public Speaker | FBLA 2023 – 2025

- Earned consecutive State-Level awards by applying quantitative analysis to technical / strategic decision-making scenarios.